

PMSM, Field-Oriented Control, and Grid Inverter Integration

A technical guide for Linear Generator and inverter-systems integration work

Study Guide

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Abstract

PMSM and field-oriented-control experience is highly transferable to a grid-interactive power-conversion role, but the controlled plant and operating objectives change. This guide develops the common inverter foundation, then translates motor-drive concepts into grid-following and grid-forming inverter control, protection, validation, and field diagnostics. Product-specific Linear Generator internals are proprietary; Mainspring public material describes pressure-driven oscillators, coils, and power electronics at a system level.

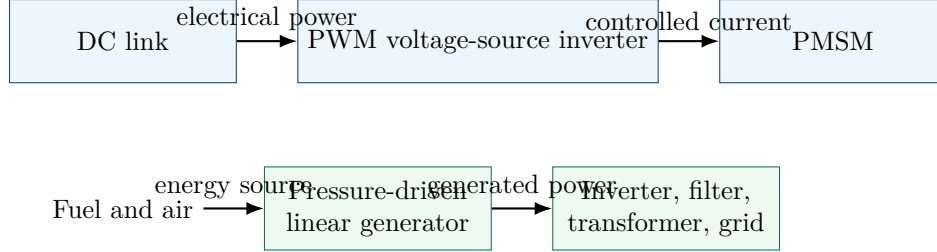
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1 Where PMSM Knowledge Transfers

1.1 The correct comparison

A PMSM converts electrical energy into rotary mechanical energy. A linear generator converts externally supplied reciprocating mechanical motion into electrical energy. Both may use permanent magnets, copper windings, position-dependent flux linkage, and PWM power electronics, but they are not the same machine.



The transferable control stack is: calibrated sensing, synchronous reference frame, current regulation, PWM, timing discipline, saturation handling, fault protection, and test evidence. The major translation is from *torque and speed* to *real/reactive power, voltage, frequency, and grid-code behavior*.

PMSM drive concept	Grid-following inverter analogue	Engineering implication
Rotor electrical angle	Grid angle estimated by PLL	Bad angle estimation corrupts dq currents.
Torque command	Active-power current command	Current limit bounds delivered power.
Speed outer loop	DC-bus voltage or active-power outer loop	Slower than the current loop.
Machine inductance	Filter plus grid impedance	Varies with cable, transformer, and grid.
Mechanical load step	Grid sag, frequency event, or site-load step	Requires ride-through and stable limiting.
Back-EMF	Grid voltage	The inverter must synthesize a voltage relative to it.

2 Three-Phase Inverter Hardware

2.1 Voltage-source inverter and PWM

A two-level, three-phase voltage-source inverter (VSI) uses six controlled switches and antiparallel current paths to connect each phase to the positive or negative DC rail. Pulse-width modulation makes the average phase voltage follow a commanded sinusoidal reference. For a sinusoidal PWM approximation in the linear modulation region,

$$V_{ph,1,peak} \approx \frac{m_a V_{dc}}{2}, \quad 0 \leq m_a \leq 1,$$

where m_a is modulation index. Space-vector PWM uses the same physical switching states but allocates zero-vector time more effectively and commonly provides better DC-bus utilization.

Dead time protects against shoot-through, yet distorts current near zero crossing. It must be characterized or compensated using current polarity, switch timing, and measured waveform evidence.

2.2 DC link, switching losses, and thermal path

The DC link buffers the mismatch between source power and AC-side power:

$$C_{dc} \frac{dV_{dc}}{dt} = \frac{P_{src} - P_{ac} - P_{loss}}{V_{dc}}.$$

Its capacitor must tolerate voltage, ripple current, temperature, expected life, and transients. Precharge limits inrush into this capacitance before a main contactor closes.

An approximate semiconductor loss model is

$$P_{loss} = P_{cond} + P_{sw}, \quad P_{sw} \approx f_s(E_{on} + E_{off}),$$

where switching energies depend on current, voltage, device temperature, and gate drive. Junction temperature follows the complete thermal chain:

$$T_j = T_a + P_{loss}(R_{\theta jc} + R_{\theta cs} + R_{\theta sa}).$$

IGBTs often favor high-power moderate-switching-frequency designs; SiC MOSFETs can reduce switching loss and enable higher frequency, but layout, gate drive, insulation stress, and EMI must be designed accordingly.

Worked example: DC-bus current

For a 250 kW converter supplied from a 1000 V DC bus at 97% converter efficiency,

$$I_{dc} \approx \frac{250000}{0.97(1000)} = 258 \text{ A}.$$

This is only the average value. Capacitor ripple-current capability, transient current, fault interruption, and busbar inductance remain separate requirements.

2.3 Output filters and sensing

An L filter limits switching-current ripple. An LCL filter reduces high-frequency harmonic injection more strongly, but its resonance must be damped and kept away from control bandwidth and uncertain grid resonances. A first estimate is

$$f_r = \frac{1}{2\pi} \sqrt{\frac{L_1 + L_2}{L_1 L_2 C_f}}.$$

Phase-current sensors require polarity checks, offset removal, gain calibration, adequate bandwidth, saturation margin, and a defined failure response. A sign error can turn negative feedback into positive feedback; treat first energization as a controlled, low-power verification rather than a production run.

3 PMSM Model and Field-Oriented Control

3.1 Reference-frame transformations

Balanced three-phase currents can be represented as a stationary two-axis vector through the Clarke transform:

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & \sqrt{3}/2 & -\sqrt{3}/2 \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}.$$

The Park transform rotates that vector by electrical angle θ_e :

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \begin{bmatrix} \cos \theta_e & \sin \theta_e \\ -\sin \theta_e & \cos \theta_e \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}.$$

For a correctly aligned frame, sinusoidal steady-state currents become nearly DC quantities. PI regulators can then track current commands without trying to regulate a sinusoid directly.

3.2 PMSM dq equations and torque

For a PMSM with electrical speed ω_e ,

$$\begin{aligned} v_d &= R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q, \\ v_q &= R_s i_q + L_q \frac{di_q}{dt} + \omega_e (L_d i_d + \lambda_m). \end{aligned}$$

The cross-coupling terms motivate feedforward decoupling. Electromagnetic torque is

$$T_e = \frac{3}{2} p [\lambda_m i_q + (L_d - L_q) i_d i_q],$$

where p is pole-pair count and λ_m is PM flux linkage. Surface-PM machines commonly use $i_d \approx 0$ below base speed; salient interior-PM machines can exploit negative i_d for maximum torque per ampere (MTPA). Above base speed, back-EMF consumes voltage headroom. Field weakening applies negative i_d so the voltage magnitude remains feasible:

$$v_d^2 + v_q^2 \leq V_{max}^2.$$

Current is separately constrained:

$$i_d^2 + i_q^2 \leq I_{max}^2.$$

These two circles are a useful mental model for every limited inverter, including a grid converter during a voltage event.

3.3 Current-loop implementation

The usual hierarchy is a fast d/q current loop inside a slower torque or speed loop. A PI law may be written as

$$v_d^* = K_{pd}(i_d^* - i_d) + K_{id} \int (i_d^* - i_d) dt + v_{d,ff},$$

with an equivalent q axis equation. Actual behavior depends on ADC sampling instant, computation delay, PWM update, sensor filtering, voltage saturation, and anti-windup. A clean simulation without these effects is not enough to establish hardware stability.

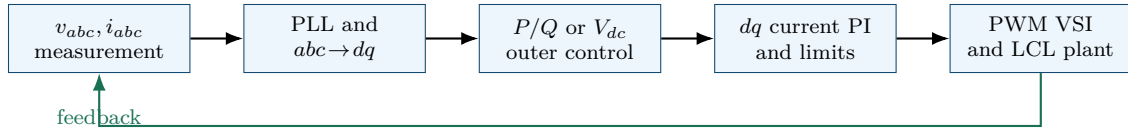
4 From FOC to Grid-Following Inverters

4.1 PLL and dq power control

A grid-following inverter uses a phase-locked loop (PLL) to estimate grid angle. With the d axis aligned to grid voltage, $v_q \approx 0$, and three-phase power becomes

$$P = \frac{3}{2}v_d i_d, \quad Q = -\frac{3}{2}v_d i_q,$$

under this stated sign convention. Different firmware may define axes or reactive-power signs differently; verify conventions in its interface control document and test data.



PLL bandwidth is a tradeoff. A fast PLL tracks disturbances quickly but may interact with a weak grid; a slow PLL is more selective but may delay synchronization. Grid strength is often screened with short-circuit ratio:

$$\text{SCR} = \frac{S_{sc}}{S_{rated}}, \quad S_{sc} \approx \frac{V_{LL}^2}{|Z_{th}|}.$$

It is only a screening metric, not proof of stable operation.

4.2 Current priorities during faults

At limited current, active and reactive demands cannot both be met arbitrarily:

$$i_d^2 + i_q^2 \leq I_{max}^2.$$

During a voltage sag, the plant may be required to inject reactive current for voltage support. This consumes current capacity and reduces possible active power. A controller therefore needs explicit priority logic, anti-windup, and documented behavior at the boundary.

Worked example: Voltage sag and current capability

Assume a 480 V line-line inverter rated 250 kVA. Its RMS line current limit is

$$I_{max} = \frac{250000}{\sqrt{3}(480)} = 301 \text{ A}.$$

At 0.70 pu voltage, supplying a full 301 A in-phase current allows only about 175 kW of real power. If reactive support takes part of the current circle, the real-power limit falls further. This is why “hold rated power through every sag” is physically impossible.

5 Grid-Forming Control and Plant Integration

Unlike grid-following control, grid-forming (GFM) control produces a voltage reference and does not rely on a stiff external voltage angle. A basic droop law is

$$\omega = \omega_0 - m_p(P - P^*), \quad V = V_0 - n_q(Q - Q^*).$$

Virtual impedance intentionally shapes the apparent source impedance:

$$\underline{V}^* = \underline{E}^* - (R_v + jX_v)\underline{I}.$$

It can improve current sharing and damping, but it reduces voltage stiffness and must be coordinated with physical filters, transformers, and protection.

For a Linear Generator plant, source dynamics, converter limits, transformer vector group, cable impedance, switchgear, and supervisory dispatch form one coupled system. The public Mainspring description supports a system-level view: pressure-driven oscillators generate electrical power; power electronics govern delivery; multiple cores form a modular package. Do not infer unpublished motor constants, internal control laws, or fault-current curves from this high-level description.

6 Protection, Power Quality, and Validation

6.1 Protection and ride-through

Protection must prevent equipment damage while coordinating with the site and utility. Typical functions include DC-bus over/undervoltage, overcurrent, desaturation or short-circuit detection, ground fault, isolation monitoring, over/undervoltage, over/underfrequency, anti-islanding, and emergency shutdown. Grid-code tests commonly include low/high-voltage ride-through and frequency response.

Total harmonic distortion is

$$\text{THD}_I = \frac{\sqrt{\sum_{h=2}^{\infty} I_h^2}}{I_1} \times 100\%.$$

Measure at a defined location and operating point. A power-quality meter is useful for trends, while high-bandwidth, isolated differential voltage probes and appropriate current probes are required to diagnose switching transients.

Test category	What to vary	Evidence to retain
First energization	Sensor offsets, phase sequence, polarity, precharge	Waveforms and controlled checklist
Power steps	P , Q , DC source, load, temperature	Overshoot, settling, current/voltage limits
Grid strength	Thevenin R/X , SCR, transformer/cable configuration	Stability margin and controller configuration
Ride-through	Voltage depth/duration, frequency trajectory	Current priority, trip behavior, recovery time
Power quality	Output power, grid state, switching mode	THD, unbalance, flicker, spectral plots
Communications	Modbus/CAN loss, stale data, command sequencing	State-machine and failsafe evidence

Commissioning mindset

- Establish a version-controlled settings baseline before changing gains or protection.
- Verify sensor polarity and scale at safe energy before enabling closed-loop power.
- Change one relevant variable at a time; capture synchronized commands, measurements, states,

and protection flags.

- Separate trigger from symptom in root-cause analysis. A trip code is evidence, not always the cause.
- Re-test the same requirement after a fix and screen the fleet for the same signature.

7 Focused Study Plan

1. Master the three-phase VSI, DC-link energy, PWM/SVPWM, sensing, losses, gate-drive protection, and L/LCL filters.
2. Derive Clarke/Park transforms and the PMSM dq model. Explain MTPA, field weakening, current and voltage limits, and anti-windup without relying on slides.
3. Implement or simulate a discrete-time PI current loop with a sample/PWM delay and saturation. Examine what changes when a sensor sign is wrong or grid inductance changes.
4. Move to grid-following control: PLL, P/Q outer loops, LCL damping, current priority, weak-grid instability, and LVRT behavior.
5. Study grid-forming voltage/frequency control, droop, virtual impedance, synchronization, islanding, black start, and transitions between operating modes.
6. Practice integration stories: select a converter, write pass/fail requirements, commission safely, capture waveforms, diagnose a failure, and close it with regression evidence.

Integration caution: Scope and safety

This guide is educational, not a substitute for manufacturer documentation, utility interconnection requirements, current adopted codes, a stamped design, or an approved energized-work procedure. High-power converter testing requires qualified personnel, correct PPE, rated equipment, arc-flash controls, and formal lockout/tagout practices.